[Total No. of Question -04] [Total No. of Pages - 02]

G H Raisoni International Skill Tech University, Pune

T.Y. B. Tech (CE/IT/CYSEC)

(Sem-V)

CAE II- WINTER 2026 (2024 Pattern)

Foundations of Data Science (25BTAIMDL507)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) Solve Q.1 or 2 and Q3 or 4. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1.(a)	What is data cleansing? Explain any four data cleansing operations such as handling missing values, removing duplicates, correcting inconsistent data, and handling outliers with examples.	[5]	3	2
b)	Consider the sentence: "Data Science is an important field for solving real world problems." Perform word tokenization, stemming, and lemmatization conceptually on the sentence. Explain the difference between stemming and lemmatization with suitable examples.	[5]	3	4
	OR			
Q 2.(a)	Explain JSON data processing with a suitable example. Write Python code to read a JSON file containing student records and display the names and marks of all students.	[5]	3	3
(b)	A CSV file named `students.csv` contains the columns Student_ID, Name, Age, Marks, and Attendance. Write Python/Pandas statements to (a) read the CSV file, (b) display the first five records, (c) find missing values, and (d) save the processed data into a new CSV file.	[5]	3	3
Q 3.(a)	A sample has a mean of 70, population standard deviation of 10, and sample size of 100. Calculate the 95% confidence interval for the population mean using z=1.96. Interpret the result.	[5]	4	3
(b)	A fair coin is tossed 5 times. Using the binomial distribution, calculate the probability of obtaining exactly 3 heads. Clearly identify nn, xx, and pp in the calculation.	[5]	4	3
	OR			
Q. 4 (a)	Calculate the population variance and standard deviation for the data: 2, 4, 6, 8, 10. Explain what the obtained variance indicates about the spread of the data.	[5]	4	3

(b)	The marks of five students are 60, 70, 80, 90, and 100. Calculate the sample mean and explain the difference between an estimator and an estimate using this example.	[5]	4	3